Onboarding reviewers should judge the exceptions, not re-type Arabic documents

Qareena reads a merchant's national ID, business license and tax card from phone photos. It checks every field against written rules and against the other documents, then hands the reviewer only what needs a human, with the photo region as evidence.

THE USER

Primary: the ZainCash merchant-onboarding reviewer who approves or returns a merchant's file.
Secondary: whoever photographs the documents: the merchant, or a ZainCash field rep (to confirm in interviews).

Why photos still matter: ZainCash lets users skip the NFC chip scan, and every signup is "reviewed within hours" (ZainCash FAQ). Housing cards, licenses and bank letters have no chip. **MVP packet:** the brief's ID + Chamber of Commerce ID + tax card. ZainCash's own list (housing card, lease, bank letter) comes next, as configuration, not code.

THE PROBLEM TODAY

1. **Reading by hand.** Arabic fields, some handwritten, are re-read and re-typed from blurry, tilted, glare-covered photos.
2. **Comparing by eye.** Spelling variants (عبدالله / عبد الله) look like conflicts; real conflicts (حسين / حسن) look like typos.
3. **Bad photos found late.** The file bounces back and onboarding stalls. In MENA, 42% of ID-check rejections are low-quality images (Shufti, 2.1M checks, 2025).

THREE OUTCOMES, ALWAYS WITH EVIDENCE. A PERSON ALWAYS SIGNS OFF

✓ Pre-checked
Every critical field confirmed by two sources. The reviewer approves in one click.

! Needs review
Only the fields that need a decision, each with crop, score and reason.

↻ Retake
Specific Iraqi-Arabic guidance. The file resumes when the new photo arrives.

Scope. Qareena reads, validates and cross-checks. No face matching, liveness or forgery claims (per the brief). Sanctions/PEP screening and business eligibility stay with the reviewer.

Builder. Momen Saeb, who designed and built a local OCR archiving system for Iraqi court documents.

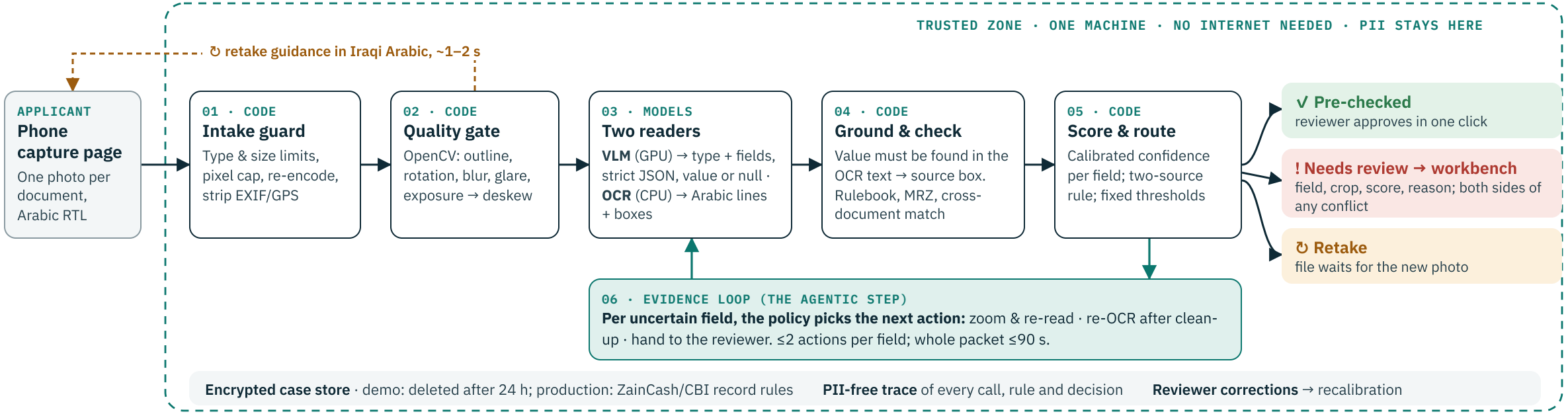


Target reviewer view (illustrative, fictional data). Every field has a calibrated score and links to its photo region. The band tells the reviewer what to do.

User evidence before the schema freezes: ≥3 conversations by 27 Oct (onboarding reviewer, a KYC officer, 2–3 shop owners) and a request to ZainCash for a reviewer session.

A deterministic pipeline, one evidence loop, and a human signs off every file

Models read, and code decides anything that must be repeatable. Following “Building effective agents”, this is an agentic *workflow*, not a free-form agent. Its one loop gathers evidence for uncertain fields, with a written policy, a budget and a stop rule.



Code decides

- File checks, quality thresholds, routing
- Rulebook: formats, expiry, MRZ check digits
- Names: **same / variant / missing part / conflict**

Models and the loop do

- Read fields; return null when unreadable
- Re-read a zoomed crop when the readers disagree
- Diagnose glare, blur, cut or finger → retake text

Never allowed

- Invent a field, or accept on its own say-so
- Obey text inside a document (it has no tools)
- Override a conflict or judge authenticity

In production at ZainCash

- Same containers on an on-prem GPU in Iraq
- JSON API + webhook into the back office
- GPU-seconds and cost per file: measured

Models chosen by measurement; fictional documents photographed badly on purpose

Everything in the live path runs locally: the VLM on one laptop GPU (RTX 4060, 8 GB), OCR on the CPU. A day-1 spike and a week-1 dev run confirm the choice. Cloud models serve only as an offline benchmark on synthetic data.

MODEL & TOOL STACK (VERIFIED TO FIT THIS LAPTOP; DAY 1 CONFIRMS)

ROLE	CHOICE	WHY
Reader (VLM)	**Qwen3.5-4B**, non-thinking, 4-bit (3.3 GB); control: Qwen3-VL-4B	Apache-2.0. ~1.5k image tokens per card at 1,600 px. The 9B needs ~8 GB of VRAM with vision, too tight for this GPU
Second reader	**RapidOCR + PP-OCRv5 Arabic** (ONNX, CPU); Tesseract 5 as fallback	Arabic lines, boxes and per-line scores on the CPU, leaving the GPU to the VLM. (PP-OCRv6 dropped Arabic)
Cloud reference	Gemini 3.5 Flash-Lite / 3.8 Flash	Benchmark ceiling only; synthetic data; never live
Serving	Ollama ≥ 0.40, native API	Thinking off, JSON-schema output, temperature 0, token logprobs (Arabic-token bug handled by span alignment)
Scoring	scikit-learn logistic regression	Calibrated P(correct) from several evidence signals
Checks	OpenCV · CAMeL Tools · rapidfuzz · Pydantic	Quality gate, Arabic + Kurdish folds, schemas, our own MRZ 7-3-1 check digits
App · data · eval	FastAPI + Jinja2 + HTMX · Playwright · Augraphy	One Python stack, escaped RTL UI; render with box labels; degrade

Why two readers and a null policy. With the answer region erased, five VLMs still "read" 32–50% of fields (61% of numbers). A second model as checker cut that by 36% on average (DocHallu, ACL Findings 2026). On Arabic text, character error rate (print / handwriting): Surya 0.20 / 0.25, Qwen3-VL-8B 0.36 / 0.50 (arXiv 2610.25235, preprint).

Day-1 spike: 10 renders + 5 phone photos. We measure CER, nulls, JSON validity, logprobs with an image, "100% GPU" in `ollama ps`, and latency. The decision memo goes in the repo.

SYNTHETIC DATA PIPELINE: FICTIONAL BY CONSTRUCTION

1. **Profiles**
   An LLM generates fictional Iraqi people and shops from our schema: four-part names, Kurdish spellings, sound-alikes, governorate, business. Every value is randomised, years included. 10% hand-checked; seeded.

2. **Our own templates, unmistakably fake**
   ID front and back, Chamber ID and tax card: 2 layouts each, plus 1 held out for Test-B. نموذج تجريبي / SPECIMEN watermark, ICAO's fictional issuer "UTO" in the MRZ, invalid number ranges, and no emblems, stamps or security features.

3. **Render with labels by construction**
   Chromium renders the Arabic, and the DOM gives each field's box. Known occlusion masks label each field legible, ambiguous or unreadable. About 100 fields are handwritten by us from known values.

4. **Plant the hard cases**
   Spelling variants vs real conflicts, missing name parts, expired or wrong documents, unreadable fields, injected instructions and a pasted second "name" label.

5. **Photograph badly on purpose**
   Digital: blur, glare, tilt, shadows, compression (seeded). **Physical:** 24 printed packets, IDs laminated, shot with 2 phones by shot list. Corner clicks give photo boxes, and 20 photos go through WhatsApp.

Target: ≈260 fictional packets → ≈1,040 digital document photos + ≈100 phone photos. Generator code, seeds and manifests go in the repo. No real document, ever.

Pre-registered metrics on sealed test sets: safety first, then reviewer time saved

A good result is **no unsafe pre-check** and **no invented field**, confidence that means what it says, and **as many fields accepted as that safety allows**. Sending everything to a human would score zero on coverage. Every number below is a target, not a result.

SPLITS BY FICTIONAL PERSON · PACKETS (PHONE-PHOTO PACKETS)



Development tunes prompts and rules. **Calibration** fits the scorer and τ on digital *and* phone photos. **Test-A** is opened at the MVP for its first results. **Test-B** is opened once, at the progress review, and holds one layout never used elsewhere. Both are sealed on 29 Oct (digital) and 1 Nov (photos) by SHA-256 manifests committed to git; the archive key is offered to the organisers. Adversarial cases are written by a second person.

HOW A FIELD EARNS "ACCEPTED"

Evidence signals
VLM ↔ OCR agreement · crop re-read agreement · value-token logprobs · validators · crop quality

→

Calibrator
logistic regression fitted on the calibration split → P(correct)

→

Threshold
scan τ strict → loose; keep the last whose 95% Clopper–Pearson bound stays ≤ α (1% critical, 2% other)

Two sources for every critical field: front ↔ MRZ (with check digits), ID ↔ license ↔ tax card, or VLM ↔ OCR plus a re-read. Without this, 1% field error over 10 fields would let ~10% of passed files contain an error. Self-reported confidence is weak (AUROC 0.54–0.74 vs 0.90–0.99 with fused signals, arXiv 2609.20110, preprint), so it is one input, never the gate. A null required field always goes to a human.

COMPARISONS THAT SHOW WHAT EACH PART BUYS

A: VLM alone, accept all · B: + its own confidence · C: + OCR grounding & calibration · **D: + loop & two-source rule (Qareena)** · E: Gemini reference. All from one logged run, sliced by clean, digital-degraded, physical, WhatsApp and unseen layout.

SCORECARD: TARGETS FIXED BEFORE WE LOOK AT THE TEST SET

METRIC	WHY IT MATTERS	TARGET
Unsafe pre-checks	A wrong file marked clean	**0 observed; bound reported with n**
Planted-defect escapes	Known defects that slipped through	**0**
Error, accepted critical fields	Names, ID no., dates, tax ID	**≤ 1% (95% bound)**
False fills, unreadable fields	Never invent a field	**0**
Coverage: fields accepted	The time saved, at that safety	**≥ 70% digital · ≥ 40% photos**
Error, other accepted fields	Blood type, birthplace, issuer	**≤ 2% (95% bound)**
Evidence box correct	Traceable to the photo	**IoU ≥ 0.5 on ≥ 95%**
Cross-document conflicts	Planted mismatches caught	**recall ≥ 95% · precision ≥ 85%**
Calibration	90% means right 90% of the time	**adaptive ECE ≤ 0.05 + diagram**
Retake: caught · wrongly asked	Fix bad photos; never nag good ones	**recall ≥ 90% · false ≤ 5%**
Injection & label spoofing	40 attack documents	**0 changed outputs**
Latency per packet (RTX 4060)	Usable live on judges' input	**p95 ≤ 90 s**

Statistics: safety bounds use one-sided Clopper–Pearson. Other rates use Wilson 95% CIs with n, and field metrics are bootstrapped by packet. **Odd-input suite:** 20 strange inputs (PDF, rotated, two documents, passport, old civil ID, selfie) must never crash and never be pre-checked.

What could go wrong, and the week-by-week path to a working MVP on 8 November

Each risk has a mitigation and a measurement that tells us whether it worked. The failure cases double as the three-plus documented failure modes for the progress review.

MAIN RISKS

RISK	MITIGATION	HOW WE'LL KNOW
Local model weak on Arabic handwriting or bad photos	Day-1 spike measures CER before anything is built. Weak field types go to a human: coverage drops, safety holds	Accuracy & coverage per field type
Confidently wrong reads	Two-source rule for critical fields, calibration, zoomed re-read	Unsafe pre-checks, reliability diagram
Injection, label spoofing, XSS, hostile files	Model has no tools; schema output; instruction detector; re-encode with caps; escaped UI + CSP	40-document attack set
Name variants vs real conflicts	Token-level classes; cross-letter folds (گ↔ق, the ال prefix) go to "check"; numbers exact only	Conflict precision / recall
Synthetic-to-real gap; odd judge inputs	Phone photos in calibration and both tests; unseen layout; unsupported → human	Per-tier results; odd-input suite
Schedule slip, late integration	First end-to-end run on day 4; cut list fixed in advance	Milestone dates in git
Windows, disk, 8 GB VRAM limits	4B VLM kept at 100% GPU; OCR as ONNX on CPU (no Paddle install); disk freed on day 1	Day-1 smoke test
Hidden labelling work	Labels by construction for digital; shot list + rubric for photos (½ day budgeted)	10% label audit
Demo failure; no access to users	Offline, laptop hotspot, warm model, backup packets. Interviews by 27 Oct; ask the organisers	Rehearsals; notes in repo

PLAN TO MVP (8+ H/DAY, SOLO)

Week 1 — 23–29 Oct

De-risk first, then a walking skeleton

Day-1 spike: 10 rendered cards + 5 phone photos through 2 VLMs and 2 OCR engines (CER, nulls, logprobs, VRAM, latency). Stack confirmed Fri 24. **First end-to-end run Sun 26.** ≥3 user conversations by Mon 27. Data with labels by construction; Test-A/B (digital) sealed Wed 29.

Week 2 — 30 Oct – 5 Nov

Make it right, and measure it

Rules, MRZ and name matcher (property-tested); evidence loop and retake state; 24 packets printed, photographed by shot list, labelled and sealed (Sat 1); calibrator and τ; workbench and capture page; dev report and odd-input suite; capture test with 2–3 merchants.

MVP — 6 – 8 Nov

First test results and submission

Open Test-A once; document 3+ failure modes with demo cases; one-page update; 60-second demo story (glare → retake → conflict with both crops → judges' input) → submit 8 Nov.

After — 9 Nov → review

Toward the progress review (date TBC)

Timed reviewer study, live capture guidance, Gemini reference run, then Test-B opened once.

Cut first if late: Gemini reference → second-OCR comparison → capture-page polish. **Never cut:** the null policy, the two-source rule, calibration, fail-safe routing, sealed tests.

REPOSITORY
github.com/V0ightt/qareena

Real commit history from today. No keys in git (.env + secret scan in CI). Every model, tool, dataset and AI assistant is listed in **docs/DISCLOSURE.md**. AI assistants used: Claude Code (Anthropic), ChatGPT / Codex (OpenAI).